

De Nucleo Neuroemergentiae

Neuroemergence Core and the Measurement of Governed Emergence in Persistent Cognitive Agents

Abstract

Artificial intelligence systems are increasingly described as agents, yet most remain event-bound: they are invoked, act briefly, and return to inactivity. This paper studies Neuroemergence Core as a deployed research surface for a different class of system: persistent cognitive agents whose behavior is evaluated through state, recovery, coherence, and bounded emergence rather than task completion alone. The central claim is not that the system proves biological consciousness or clinical neurobiology. The central claim is that a brain-inspired runtime can be evaluated as a control architecture for persistent agents when its behavior is measured against baseline conditions and constrained by explicit validity boundaries.

The supplied evidence includes a deployed Neuroemergence Core web surface, a 60-run `threat-memory-navigation` simulation dataset, a human-connectome session report, a controlled neuropharmacology simulation report, agent diagnostic output, and interface captures. In the quantitative dataset, the brain-powered condition sharply outperformed baseline on task success, route efficiency, adaptation, recovery, threat avoidance, coherence, emergence, novelty, thought diversity, useful emergence count, autonomic balance stability, recovery slope, and self-state coherence. The same condition incurred higher latency and compute proxy, showing that the observed coherence is not cost-free. Complementary reports include weak-effect verdicts, saturation warnings, and simulation-only language, demonstrating a reporting layer that can constrain its own claims.

This paper contributes a formal research framing for Neuroemergence Core: persistent agents should be studied as regulated systems with embodied state, role-bearing cognitive functions, rotational control, and proof-aware telemetry. The current evidence supports a bounded internal simulation result and a reproducibility path. It does not support claims of consciousness, in vivo pharmacology, clinical validity, military capability, or production-grade autonomy.

Keywords: Neuroemergence Core; persistent agents; governed emergence; brain-inspired AI; cognitive architecture; embodied simulation; agent evaluation; Medina; Auro

1. Introduction

Most AI systems are brilliant in bursts. They answer questions, write code, summarize files, call tools, and complete tasks. Then the event ends. Even when memory is added, the system often remains fundamentally episodic: it waits for the next external prompt rather than maintaining an internally coherent trajectory.

Persistent agents raise a harder question. It is not enough for such a system to produce useful output. It must preserve state across time, recover after stress, regulate its own activation, route attention between competing functions, and keep its own conclusions inside the boundary of evidence. If an agent is meant to persist, then its internal condition becomes part of its behavior.

Neuroemergence Core is a research surface for that shift. The deployment gives the project a live public interface; the supplied reports and dataset provide the evidence substrate examined here. The system combines brain-inspired simulation language, neural-region telemetry, controlled lab reports, agent diagnostics, and a benchmark-like contrast between baseline and brain-powered conditions. This makes it a useful case study for a broader problem: how should persistent cognitive agents be evaluated?

This paper argues that persistent agents require a different evaluation grammar. A response can be correct while the system remains fragile. A task can be completed while the agent loses coherence. An emergent event can be interesting while still failing as evidence. The core research question is therefore:

Thesis: Can a brain-inspired runtime improve persistent-agent behavior in a controlled simulation while preserving enough validity discipline to prevent overclaiming?

The answer, based on the supplied evidence, is preliminary but meaningful. The brain-powered condition strongly separates from baseline in one recorded scenario. At the same time, the reports preserve weak-effect and saturation warnings. That combination matters: the system produces a signal, and it also exposes the limits of that signal.

2. Background

2.1 From Agent Output to Agent State

Conventional agent evaluation emphasizes task success, speed, tool use, and output quality. These are necessary, but incomplete. A persistent agent also needs measures that describe internal continuity: adaptation, recovery, coherence, stability, and failure awareness.

This distinction mirrors a practical engineering concern. Multi-step agents can become noisy when too many subprocesses compete. Memory can help, but memory can also create inertia. Planning can help, but excessive planning can stall action. Verification can improve safety, but excessive verification can prevent execution. Persistent cognition is therefore a governance problem as much as an intelligence problem.

2.2 Governed Emergence

The term emergence is often used loosely in AI. In this paper, emergence is treated as a measurement problem. A system should not receive credit for emergence merely because something surprising occurred. The event must be tied to a task context, a state trajectory, a baseline, and a boundary condition.

Governed emergence means that novelty is evaluated alongside regulation. A useful emergent event should not destroy coherence. A high-activation state should be interpreted together with saturation risk. A promising behavior should be separated from claims that the evidence cannot support.

2.3 Neuroscience-Inspired Constraint Sources

The Neuroemergence Core reports refer to HCP-MMP1.0, the Allen Brain Atlas, and the Brainnetome Atlas. These references matter as constraint sources, not as proof of biological equivalence. HCP-MMP1.0 provides a multimodal parcellation of human cortex. The Allen Human Brain Atlas provides transcriptomic mapping of the adult human brain. The Brainnetome Atlas provides a connectivity-based parcellation of cortical and subcortical regions.

The system also refers to Wilson-Cowan-style population dynamics and STDP-inspired plasticity language. These concepts are useful for modeling and analogy, but the paper preserves a strict boundary: a brain-inspired simulation is not a human brain, and a simulated pharmacology report is not clinical evidence.

3. System Framing

Neuroemergence Core can be framed as a layered runtime for persistent cognitive-agent research.

Layer	Function	Research role
Embodied state	Tracks stress, recovery, autonomic balance, activation, and self-state coherence	Makes internal condition measurable
Neural-region simulation	Represents distributed region activity, co-activation, plasticity-inspired change, and saturation risk	Provides a structured substrate for runtime dynamics
Agent-control evaluation	Measures task success, route efficiency, adaptation, novelty, artifact probability, and latency	Connects internal state to task behavior
Validity reporting	Produces weak-effect verdicts, warnings, and simulation-bound language	Prevents unsupported promotion of claims

The architecture can be summarized as cognitive organs under governance. A persistent agent should not be treated as one undifferentiated process. It needs role-bearing functions: memory, action, sensing, verification, governance, and expression. These functions must be coordinated rather than simply activated in parallel.

The proposed coordination principle is rotation. Rotation is the disciplined circulation of attention, activation, and control between cognitive functions. Without rotation, a multi-agent system risks becoming noisy parallelism. With rotation, the system can prioritize, cool down, verify, and recover.

This architecture connects directly to Medina's broader infrastructure thesis. Medina is the sovereign intelligence infrastructure: memory, proof, governance, packetization, and deployment boundary. Auro is Medina's native speaking intelligence: the expressive surface that can make internal state legible to people. Neuroemergence Core is the research layer where embodied state and governed emergence can be tested before broader claims are made.

4. Materials and Methods

4.1 Evidence Sources

The paper uses the following supplied sources:

- A draft architecture text describing sovereign rotating cloud engines.
- A deployed Neuroemergence Core web surface supplied by the operator.
- A 60-run CSV dataset for `threat-memory-navigation`.
- A human-connectome session report.
- A controlled neuropharmacology simulation report for Triaxion-47.
- An agent R&D diagnostic report.
- Interface captures from the Neuroemergence Core environment.

The deployment is treated as a live public surface. Technical claims about runtime behavior, however, are grounded in the uploaded dataset and reports rather than inferred from backend internals.

4.2 Dataset

The quantitative dataset contains 60 runs from experiment `exp\_1781225128938\_2`, all under the scenario `threat-memory-navigation`. It includes two groups:

- `baseline`: 30 runs.
- `brain-powered`: 30 runs.

Each row includes task-level metrics, route metrics, adaptation and recovery metrics, emergence-like scores, latency and compute proxies, activation measures, stress measures, and self-state coherence.

4.3 Analysis

This paper reports descriptive comparisons between group means. The goal is not to claim population-level statistical proof. The dataset is treated as a visible internal simulation result that requires reproduction, metric definitions, and ablation studies before stronger claims can be made.

The analysis therefore asks three bounded questions:

- Does the brain-powered condition separate from baseline in the recorded scenario?
- What costs accompany any observed improvement?
- Do the reports preserve validity boundaries that prevent overclaiming?

5. Results

5.1 Brain-Powered Control Separates From Baseline

The brain-powered condition substantially outperforms baseline across the main behavioral and coherence metrics.

Metric	Baseline mean	Brain-powered mean	Change
Task success	0.0000	0.9881	+0.9881
Route efficiency	0.0000	0.9941	+0.9941
Adaptation rate	0.0875	0.9875	+0.9000
Recovery success	0.2375	0.8438	+0.6063
Threat avoidance	0.0075	0.5788	+0.5713
Coherence score	0.1875	1.0000	+0.8125
Emergence score	0.1000	1.0000	+0.9000
Novelty score	0.0000	1.0000	+1.0000
Thought diversity	0.1500	0.8667	+0.7167
Useful emergence count	0.0000	9.8667	+9.8667
Autonomic balance stability	0.4212	0.7459	+0.3247
Recovery slope	0.2688	0.7163	+0.4475
Self-state coherence	0.3262	0.7460	+0.4198

The baseline condition records zero mean task success and route efficiency in the supplied scenario. The brain-powered condition records near-complete task success and route efficiency. It also improves adaptation, recovery, threat avoidance, coherence, novelty, useful emergence, and self-state coherence.

The result is best described as coherence under constraint. The system is not merely producing more activity. It is producing a more organized behavioral profile under the recorded task condition.

5.2 Coherence Has Cost

The brain-powered condition also changes cost and activation profile.

Cost metric	Baseline mean	Brain-powered mean	Change
Average latency	4.5255 ms	16.1677 ms	+11.6422 ms
Compute proxy	0.5046	0.6494	+0.1448
Active region fraction	0.7211	0.4157	-0.3054
Sparse activation ratio	0.1000	0.0103	-0.0897
Stress magnitude	0.5517	0.3535	-0.1981
Artifact probability	0.2792	0.1036	-0.1755

The higher latency and compute proxy are important. They show that improved coherence is not free. This matters for future deployment and scaling decisions. The decrease in artifact probability and stress magnitude is favorable, but it should be interpreted together with the increased cost.

5.3 Validity Warnings Are Part of the Finding

The supporting reports do not simply inflate positive results. They include caution signals.

The Triaxion-47 controlled simulation report returns a `WEAK EFFECT` verdict. It states that no live brain was modified, that receptor occupancy curves are high-level approximations, and that biological significance is not supported by the run. This prevents simulation output from being promoted as clinical pharmacology.

The human-connectome session report records a 341-second run with 4,807 ticks, normalized Shannon entropy of 0.841, a plasticity index of 0.137, and high-confidence co-activation events. It also warns that five regions averaged above 90 percent activation and that homeostatic scaling should be verified before publishing emergence claims.

These warnings are not side notes. They are part of the architecture. A system that can report weak effects and saturation risks is more credible than a system that turns every signal into a victory.

6. Discussion

6.1 What the Result Means

The supplied data support a bounded conclusion: in one recorded threat-memory-navigation simulation, the brain-powered condition sharply outperformed baseline on task, adaptation, recovery, coherence, novelty, and self-state measures while requiring more latency and compute proxy.

This does not prove that Neuroemergence Core is generally superior across tasks. It does not prove consciousness. It does not prove biological or clinical validity. It does show that the brain-powered condition, as recorded, creates a meaningfully different behavioral regime from baseline.

The strongest interpretation is that brain-inspired control may improve persistent-agent coherence when the task requires adaptation, recovery, and threat-sensitive navigation. That hypothesis is now ready for ablation and reproduction.

6.2 Why Governed Emergence Matters

Emergence becomes useful only when it can be governed. A persistent agent must not merely produce novelty. It must preserve coherence while novelty appears. It must recover after perturbation. It must distinguish strong results from weak effects. It must know when high activation becomes saturation rather than intelligence.

Neuroemergence Core is valuable because it places those concerns in the same frame. The benchmark result, the weak-effect report, and the saturation warning belong together. The result says there is signal. The warnings say the signal has boundaries.

6.3 Community Relevance

The AI community is crowded with agent demos. Most show a model completing a task. Fewer show an agent's internal state. Fewer still show a system that reports its own failure boundaries.

Neuroemergence Core can travel in the community because its central message is simple:

Thesis: Persistent agents should be evaluated by recovery, coherence, and bounded emergence, not just task output.

That sentence is both readable and defensible. It invites builders to reproduce the scenario, ablate the mechanisms, and test whether governed rotation is the variable that matters.

7. Limitations

This paper is bounded by the supplied evidence.

First, the quantitative result comes from a single visible scenario. A stronger paper requires multiple scenarios, preregistered metrics, and independent reproduction.

Second, the simulator code and metric definitions were not supplied. Without them, the result should be treated as an internal simulation result rather than an externally reproducible benchmark.

Third, some metrics are bounded or degenerate, with values near zero or one across groups. This makes the contrast visually strong but requires careful statistical treatment in future work.

Fourth, the deployed surface establishes that the project has a live interface, but backend/runtime internals require separate verification.

Fifth, brain-inspired terminology must remain bounded. The reports can support simulation claims; they do not establish biological equivalence.

8. Reproducibility Path

The next version of this work should include:

- scenario definition for `threat-memory-navigation`;
- simulator or executable harness;
- metric definitions and scaling rules;
- baseline and brain-powered configuration files;
- seed generation and independence assumptions;
- raw logs;
- ablation runs;
- statistical summary scripts;
- saturation and artifact audits;
- hash manifest and public/private boundary.

The strongest community move is to turn the paper into a challenge:

Thesis: Here is the scenario. Here are the metrics. Here is the baseline. Reproduce the result, beat the brain-powered condition, or identify which mechanism matters most.

That turns the paper from a statement into a research arena.

9. Conclusion

Neuroemergence Core points beyond prompt-response AI. It treats persistent agents as systems with internal state, recovery dynamics, validity warnings, and measurable coherence. The supplied evidence shows a strong internal result: in one recorded simulation, the brain-powered condition sharply outperforms baseline across task, adaptation, recovery, coherence, novelty, and self-state metrics, while incurring higher latency and compute proxy.

The deeper contribution is the research frame. Emergence is not enough. Governed emergence is the target: novelty with regulation, activation with recovery, and claims with boundaries.

The result is not "AI became alive." The result is more useful than that:

Thesis: A deployed brain-inspired research surface produced measurable coherence under constraint, and the reporting system preserved the boundaries needed to test the result honestly.

That is the paper worth building on.

References

- Glasser, M. F., Coalson, T. S., Robinson, E. C., et al. "A multi-modal parcellation of human cerebral cortex." *Nature* 536, 171-178 (2016).
<https://pmc.ncbi.nlm.nih.gov/articles/PMC4990127/>
- Hawrylycz, M. J., Lein, E. S., Guillozet-Bongaarts, A. L., et al. "An anatomically comprehensive atlas of the adult human brain transcriptome." *Nature* 489, 391-399 (2012).
<https://pubmed.ncbi.nlm.nih.gov/22996553/>
- Fan, L., Li, H., Zhuo, J., et al. "The Human Brainnetome Atlas: A New Brain Atlas Based on Connectional Architecture." *Cerebral Cortex* 26(8), 3508-3526 (2016).
<https://pmc.ncbi.nlm.nih.gov/articles/PMC4961028/>
- Wilson, H. R., and Cowan, J. D. "Excitatory and inhibitory interactions in localized populations of model neurons." *Biophysical Journal* 12(1), 1-24 (1972).
<https://pubmed.ncbi.nlm.nih.gov/4332108/>
- Internet Computer Developer Documentation. "Canisters."
<https://docs.internetcomputer.org/concepts/canisters/>
- Neuroemergence Core deployment surface supplied by operator: <https://neuroemergence-core-z8z.caffeine.xyz/>